Theory of Computation-PROBLEM SET 3

April 8, 2026

Due Apr 23rd. Upload to learn.tsinghua.edu.cn.

Problem 1

Recall that a context-free grammar G is called *ambiguous* if there is a string in $L(G)$ that has two distinct parse trees. Let $AMBIG_{CFG} = \{\langle G\rangle : G \text{ is an ambiguous CFG.}\}$. Show that $AMBIG_{CFG}$ is undecidable.

Problem 2

A language L is *self-dual* if $L \leq_m \overline{L}$. Consider some coding method $\langle M\rangle$ of Turing machines M, a language A is an *index set* if for every pair of equivalent TMs M, N, either $\langle M\rangle, \langle N\rangle$ are both in A or neither of them are in A.

(a) Show that no index set A can be self-dual.

(b) Give an example of undecidable self-dual languages.

Problem 3

In 1962, Tibor Rado defined $S(n)$, or the n-th "Busy Beaver shift number" to be the maximum number of steps made by any n-state Turing machine that eventually halts. Here, a Turing machine has a two-way infinite tape with either 0 or 1 on each square, and all tape squares are initially set to 0. (The "halting state" doesn't count as a state; it's just an operation.)

(a) Show that $S(1) = 1$.

(b) Show that $S(2) \geq 6$.

(c) Show that $S(n)$ grows faster than any computable function. In other words, there is no computable function C such that $C(n) \geq S(n)$ for all n.

(d) Show that there is not even a computable function C such that $C(n) \geq S(n)$ for infinitely many n.

Problem 4

Let $BOTH_{NFA} = \{\langle M_1, M_2\rangle : M_1 \text{ and } M_2 \text{ are NFAs where } L(M_1) \cap L(M_2) \neq \emptyset\}$. Show that $BOTH_{NFA}$ is NL-complete.

Problem 5

Recall the generalized geography game introduced in class. Consider the game played on an *undirected* graph G and starting at a vertex s.

(a) Show that the first player has a winning strategy if and only if every maximum matching of G contains s

(b) Show that the undirected GG game is in P.

(c) When the game is played on directed graphs but no repetition of edges is allowed (rather than no repetition of vertices), is it still PSPACE-complete? Prove your answer.

Problem 6 (Optional)

Show that for any c, some strings x and y exist, where $K(xy) > K(x) + K(y) + c$.